MULTIMEDIA TECHNOLOGY

Unit-2: Sound in Multimedia

Comprehensive Study Notes

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Quick Reference Summary

1. Sound and Its Attributes

Definition

Sound is a vibration that propagates as an acoustic wave through a transmission medium such as air, water, or solids. In multimedia, sound is a crucial element that enhances user experience and engagement.

Nature of Sound

Sound is created by vibrations that create pressure waves in the air. These waves are detected by our ears and interpreted by our brain.

```
+------------------------------------------------------------------+
|               PHYSICAL NATURE OF SOUND                            |
+------------------------------------------------------------------+
```

Sound Attributes Table

Attribute	Description	Measurement	Effect
Amplitude	Height of sound wave	Decibels (dB)	Determines volume/loudness
Frequency	Number of cycles per second	Hertz (Hz)	Determines pitch
Wavelength	Distance between wave peaks	Meters	Affects sound propagation
Timbre	Tone quality/color	Subjective	Distinguishes instruments
Duration	Length of sound	Seconds	Defines sound length
Phase	Wave position in cycle	Degrees	Affects sound interference
Velocity	Speed of sound wave	m/s (343 m/s in air)	Affects timing

Human Hearing Range

```
+----------------------------------------------------------------------+
|                    HUMAN HEARING RANGE                                |
+----------------------------------------------------------------------+
|                                                                       |
|   20 Hz ---------------------------------------- 20,000 Hz          |
|        |                    |                            |            |
|   Infrasound          Audible Range                Ultrasound        |
|   (Below 20 Hz)    (20 Hz - 20 kHz)             (Above 20 kHz)       |
|                                                                       |
|        +-----------------------------------------+                  |
|        |           SPEECH RANGE                  |                    |
|        |         300 Hz - 3,400 Hz                 |                    |
|        +-----------------------------------------+                  |
|                                                                       |
|        +-----------------------------------------+                  |
|        |           MUSIC RANGE                   |                    |
|        |        20 Hz - 20,000 Hz                  |                    |
|        +-----------------------------------------+                  |
+----------------------------------------------------------------------+
```

2. Mono vs Stereo Sound

Mono Sound (Monophonic)

```
+----------------------------------------------------------------------+
|                         MONO SOUND                                    |
+----------------------------------------------------------------------+
|                                                                        |
|   +------------------------------------------------------------+      |
|   |                                                            |      |
|   | ONE CHANNEL ---------------> SINGLE                       |      |
|   | (Single Audio ---------------> SPEAKER                     |      |
|   |  Source)                     |                              |      |
|   |                                                            |      |
|   | All sound mixed into one signal                            |      |
|   | No spatial separation                                      |      |
|   | Same audio from all speakers                               |      |
|   |                                                            |      |
|   +------------------------------------------------------------+      |
|                                                                        |
|   CHARACTERISTICS:                                                      |
|   * Single audio channel                                               |
|   * No left/right separation                                           |
|   * Smaller file size                                                  |
|   * Less processing power required                                     |
|   * Older technology                                                   |
|   * Still used in radio, telephones                                    |
|   * File extension: .mono                                              |
|                                                                        |
+----------------------------------------------------------------------+
```

Stereo Sound (Stereophonic)

```
+----------------------------------------------------------------------+
|                      STEREO SOUND                                     |
+----------------------------------------------------------------------+
|                                                                        |
|   +------------------------------------------------------------+      |
|   |                                                            |      |
|   | LEFT CHANNEL ---------------> LEFT SPEAKER                 |      |
|   | (Audio Source 1)              |                             |      |
|   |                                                            |      |
|   | RIGHT CHANNEL ---------------> RIGHT SPEAKER               |      |
|   | (Audio Source 2)              |                             |      |
|   |                                                            |      |
|   +------------------------------------------------------------+      |
|                                                                        |
|   CHARACTERISTICS:                                                      |
|   * Two audio channels                                                 |
|   * Left/right separation                                              |
|   * Creates spatial perception                                         |
```

```
|   * Gives directionality to sound                                      |
|   * Larger file size (approx 2x mono)                                  |
|   * More processing power required                                     |
|   * Standard for music, movies                                         |
|   * Creates immersive experience                                       |
|                                                                        |
+----------------------------------------------------------------------+
```

Mono vs Stereo Comparison

Feature	Mono Sound	Stereo Sound
Channels	1 channel	2 channels
Spatial Effect	No spatial separation	Left/Right separation
Directionality	No directional cues	Directional cues present
File Size	Smaller	Larger (2x mono)
Processing	Less intensive	More intensive
Sound Quality	Flat, centered	Rich, spacious
Equipment Needed	Single speaker	Two speakers
Best Used For	Voice, radio, phones	Music, movies, games
Cost	Less expensive	More expensive
Immersion	Low	High

3. Sound Channels

Concept of Sound Channels

```
+--------------------------------------------------------------+
|                SOUND CHANNELS CONFIGURATIONS                  |
+--------------------------------------------------------------+
```

Detailed Channel Configurations

Configuration	Channels	Speaker Layout	Application
Mono (1.0)	1	Single center	Radio, Voice
Stereo (2.0)	2	L + R	Music, TV
2.1	3	L + R + Subwoofer	PC speakers
4.1	5	Front L/R + Rear L/R + Sub	Gaming
5.1	6	Front L/R + Rear L/R + Center + Sub	Home theater
6.1	7	5.1 + Rear Center	Movies
7.1	8	5.1 + Side L/R	Advanced theater
Dolby Atmos	Variable	Multiple overhead + floor	Cinema, VR

4. Sound and Its Effect in Multimedia

Role of Sound in Multimedia

```
+----------------------------------------------------------------+
|              EFFECTS OF SOUND IN MULTIMEDIA                     |
+----------------------------------------------------------------+
```

Types of Sound in Multimedia

Type	Purpose	Examples
Background Music	Sets mood, atmosphere	Instrumental tracks
Narration/Voice-over	Explains content	Educational videos
Sound Effects (SFX)	Enhances realism	Footsteps, explosions
Ambient Sound	Creates environment	Rain, traffic, wind
Dialogues	Communication	Character speech
User Interface Sounds	Feedback	Clicks, notifications
Audio Cues	Signals actions	Alerts, warnings

Psychological Effects of Sound

```
+----------------------------------------------------------------+
|              PSYCHOLOGICAL EFFECTS OF SOUND                     |
+----------------------------------------------------------------+
|                                                                 |
| SOUND TYPE                      PSYCHOLOGICAL EFFECT            |
|-------------------------------+---------------------------------|
| Slow tempo music              | Relaxation, calmness            |
| Fast tempo music              | Excitement, energy              |
| Minor key music               | Sadness, melancholy             |
| Major key music               | Happiness, joy                  |
| Dissonant sounds              | Tension, discomfort             |
| Consonant sounds              | Harmony, comfort                |
| White noise                   | Focus, concentration            |
| Sudden loud sounds            | Alert, startled                 |
| Nature sounds                 | Peaceful, natural               |
| Voice pitch variation         | Emotion, meaning                |
```

```
|                                                                      |
+----------------------------------------------------------------------+
```

5. Analog vs Digital Sound

Analog Sound

```
+----------------------------------------------------------------------+
|                          ANALOG SOUND                                |
+----------------------------------------------------------------------+
|                                                                      |
|   +------------------------------------------------------------+    |
|   |                                                            |    |
|   | Sound Wave --> Microphone --> Analog Signal                 |    |
|   | (Continuous)  (Transducer)   (Continuous voltage)          |    |
|   |                                                            |    |
|   |              Continuous Wave Signal                        |    |
|   |              /\    /\    /\    /\                           |    |
|   |             /  \  /  \  /  \  /  \                          |    |
|   |            /    \/    \/    \/    \                         |    |
|   |                                                            |    |
|   +------------------------------------------------------------+    |
|                                                                      |
|   CHARACTERISTICS:                                                    |
|   * Continuous signal                                                |
|   * Infinite values                                                   |
|   * Natural representation                                           |
|   * Subject to noise and distortion                                  |
|   * Degrades with copying                                            |
|   * Examples: Vinyl records, cassette tapes                          |
|                                                                      |
+----------------------------------------------------------------------+
```

Digital Sound

```
+----------------------------------------------------------------------+
|                         DIGITAL SOUND                                |
+----------------------------------------------------------------------+
|                                                                      |
|   +------------------------------------------------------------+    |
|   |                                                            |    |
|   | Sound Wave --> ADC --> Digital Signal --> DAC -->         |    |
|   | (Continuous) (Sampling) (Discrete)    (Conversion)         |    |
|   |                                                            |    |
|   |              Sampled Discrete Signal                       |    |
|   |          .  .  .  .  .  .  .  .  .  .  .                 |    |
|   |        .  .  .  .  .  .  .  .  .  .  .  .  .            |    |
|   |       .  .  .  .  .  .  .  .  .  .  .  .              |    |
|   |                                                            |    |
|   +------------------------------------------------------------+    |
|                                                                      |
|   CHARACTERISTICS:                                                    |
|   * Discrete samples                                                 |
|   * Finite values (1s and 0s)                                       |
```

```
|   * Converted from analog                                         |
|   * No degradation on copying                                     |
|   * Resistant to noise                                            |
|   * Examples: CD, MP3, WAV files                                  |
|                                                                   |
+-------------------------------------------------------------------+
```

Analog vs Digital Comparison

Feature	Analog Sound	Digital Sound
Signal Type	Continuous wave	Discrete samples
Values	Infinite	Finite (binary)
Quality	Original, natural	Depends on sampling
Noise	Susceptible to noise	Noise-resistant
Copying	Degrades with copying	Perfect copies
Storage	Physical media	Electronic files
Editing	Difficult	Easy
Format	Cassette, Vinyl	WAV, MP3, CD
Cost	Generally cheaper	Higher initial cost
Durability	Fragile	Durable

ADC (Analog to Digital Conversion) Process

```
+-------------------------------------------------------------------+
|             ANALOG TO DIGITAL CONVERSION PROCESS                  |
+-------------------------------------------------------------------+
|                                                                   |
|   Analog Signal --> Sampling --> Quantization --> Encoding        |
|                        |               |               |          |
|                    +--------+      +--------+      +--------+     |
|                    |Samples |      | Assign |      | Binary |     |
|                    |   at   |      | Values |      |Numbers |     |
|                    |intervals|     |        |      |        |     |
|                    | e.g.,  |      | e.g.,  |      | e.g.,  |     |
|                    |44.1 kHz|      | 16-bit |      |01010011|     |
|                    +--------+      +--------+      +--------+     |
|                                                                   |
|   TWO KEY PARAMETERS:                                             |
|   1. Sampling Rate - How often samples are taken (Hz)             |
```

```
|   2. Bit Depth - Number of bits per sample (resolution)        |
|                                                                |
+----------------------------------------------------------------+
```

Sampling Quality Comparison

Quality	Sampling Rate	Bit Depth	Applications
Telephone	8 kHz	8-bit	Voice calls
AM Radio	11 kHz	8-bit	Radio broadcast
FM Radio	22 kHz	16-bit	Radio broadcast
CD Quality	44.1 kHz	16-bit	Music CDs
DVD Quality	48 kHz	24-bit	Movies
Studio Quality	96 kHz	24-bit	Professional recording
High Resolution	192 kHz	32-bit	Advanced mastering

6. Overview of Various Sound File Formats

Sound File Formats Classification

```
+------------------------------------------------------------------+
|              SOUND FILE FORMATS CLASSIFICATION                    |
+------------------------------------------------------------------+
```

6.1 WAV (Waveform Audio File Format)

```
+------------------------------------------------------------------+
|                       WAV FILE FORMAT                             |
+------------------------------------------------------------------+
|                                                                    |
|   FULL NAME: Waveform Audio File Format                           |
|   DEVELOPED BY: Microsoft & IBM                                   |
|   FILE EXTENSION: .wav                                            |
|   TYPE: Uncompressed                                              |
|                                                                    |
|   WAV FILE STRUCTURE:                                              |
|   +------------------------------------------------+             |
|   | HEADER (44 bytes)                               |             |
|   | - File type (RIFF)                              |             |
|   | - File size                                     |             |
|   | - Audio format                                  |             |
|   | - Sample rate (e.g., 44.1 kHz)                 |            |
|   | - Bit depth (e.g., 16-bit)                     |            |
|   | - Number of channels (mono/stereo)             |            |
|   | - Data size                                     |             |
|   +------------------------------------------------+             |
|   | DATA                                            |             |
|   | - Raw audio samples                             |             |
|   | - PCM (Pulse Code Modulation) data              |             |
|   +------------------------------------------------+             |
|                                                                    |
|   CHARACTERISTICS:                                                  |
|   * Uncompressed audio                                             |
|   * High quality                                                   |
|   * Very large file size                                           |
|   * Ideal for professional use                                     |
|   * Standard on Windows platform                                   |
|   * Widely supported                                               |
|                                                                    |
|   WAV FILE SIZE CALCULATION:                                       |
|   File Size = Sample Rate x Bit Depth x Channels x Duration        |
|   Example: CD Quality (44.1 kHz x 16-bit x 2 x 60 seconds)       |
|   = 44,100 x 2 x 2 x 60 = 10.5 MB per minute                      |
|                                                                    |
+------------------------------------------------------------------+
```

6.2 MP3 (MPEG Audio Layer 3)

```
+----------------------------------------------------------------------+
|                       MP3 FILE FORMAT                                |
+----------------------------------------------------------------------+
|                                                                      |
|   FULL NAME: MPEG-1 Audio Layer 3                                    |
|   DEVELOPED BY: Fraunhofer Society                                   |
|   FILE EXTENSION: .mp3                                               |
|   TYPE: Compressed (Lossy)                                           |
|                                                                      |
|   MP3 COMPRESSION PROCESS:                                           |
|   +-----------------------------------------------------+            |
|   | Original Audio --> Filter --> Quantize --> Bit      |            |
|   |    (WAV)       (PSYCHO-    (Reduce     (MP3)        |            |
|   |                  ACOUSTIC    Data)                  |            |
|   |                  MODEL)                             |            |
|   +-----------------------------------------------------+            |
|                                                                      |
|   Uses Psychoacoustic Model:                                         |
|   - Removes inaudible sounds                                         |
|   - Masking effect (loud sounds hide soft ones)                      |
|   - Reduces file size by 90%                                         |
|                                                                      |
|   CHARACTERISTICS:                                                   |
|   * High compression ratio                                           |
|   * Good quality at high bit rates                                   |
|   * Variable and constant bit rates                                  |
|   * Most popular format for music                                    |
|   * Universal support                                                |
|   * Small file size                                                  |
|                                                                      |
|   MP3 BIT RATES AND QUALITY:                                         |
|   +-------------------------------------------------------------+    |
|   | BIT RATE    | QUALITY      | USE CASE                       |    |
|   +-------------+--------------+--------------------------------+    |
|   | 320 kbps    | Excellent    | Audio CDs, Archiving           |    |
|   | 256 kbps    | Very Good    | High quality music             |    |
|   | 192 kbps    | Good         | Standard music                 |    |
|   | 128 kbps    | Acceptable   | Streaming, Portable            |    |
|   | 64 kbps     | Poor         | Voice only                     |    |
|   +-------------------------------------------------------------+    |
|                                                                      |
|   FILE SIZE COMPARISON (3 minutes):                                  |
|   WAV (CD Quality) = 30 MB                                           |
|   MP3 (128 kbps)   = 2.8 MB                                          |
|   MP3 (192 kbps)   = 4.2 MB                                          |
|   MP3 (320 kbps)   = 7.0 MB                                          |
|                                                                      |
+----------------------------------------------------------------------+
```

6.3 Other Sound File Formats

```
+----------------------------------------------------------------+
|              SOUND FILE FORMATS COMPARISON TABLE               |
+----------------------------------------------------------------+
+-------------------------------------------------------------+
| FORMAT | TYPE       | QUALITY   | SIZE       | USE CASE      |
+--------+-----------+-----------+------------+-----------------+
| WAV    | Uncomp.    | Excellent | Very Large | Production    |
| AIFF   | Uncomp.    | Excellent | Very Large | Mac Audio     |
| MP3    | Lossy      | Good      | Small      | Music, Web    |
| AAC    | Lossy      | Better    | Small      | iTunes, Web   |
| OGG    | Lossy      | Good      | Small      | Open Source   |
| WMA    | Lossy      | Good      | Small      | Windows       |
| FLAC   | Lossless   | Excellent | Medium     | Archiving     |
| ALAC   | Lossless   | Excellent | Medium     | Apple Music   |
| PCM    | Uncomp.    | Excellent | Very Large | Recording     |
| MIDI   | Control    | Depends   | Very Small | Synthesizer   |
+-------------------------------------------------------------+
```

Detailed Format Descriptions

Format	Description	Advantages	Disadvantages
WAV	Windows standard format	High quality, no loss	Large file size
AIFF	Apple standard format	High quality, Mac native	Large file size
MP3	Most popular compressed format	Small size, universal	Lossy compression
AAC	Advanced Audio Coding	Better quality than MP3	Less universal
OGG	Open source format	Free, good quality	Limited support
WMA	Windows Media Audio	Small size, Windows support	Platform dependent
FLAC	Free Lossless Audio Codec	Lossless, smaller than WAV	Less universal
ALAC	Apple Lossless	Lossless, Apple ecosystem	Limited to Apple devices
MIDI	Musical Instrument Digital Interface	Extremely small, editable	Not actual audio

6.4 MIDI (Musical Instrument Digital Interface)

```
+----------------------------------------------------------------------+
|                        MIDI FILE FORMAT                              |
+----------------------------------------------------------------------+
|                                                                      |
|   FULL NAME: Musical Instrument Digital Interface                   |
|   FILE EXTENSION: .mid, .midi                                       |
|   TYPE: Control/Instruction format                                  |
|                                                                      |
|   MIDI CHARACTERISTICS:                                              |
|   +-----------------------------------------------------+           |
|   |                                                     |           |
|   |  * Does NOT contain audio data                      |           |
|   |  * Contains instructions for instruments            |           |
|   |  * Data includes:                                   |           |
|   |    - Note On/Off events                             |           |
|   |    - Pitch                                          |           |
|   |    - Velocity (loudness)                            |           |
|   |    - Instrument selection                           |           |
|   |    - Duration                                       |           |
|   |                                                     |           |
|   +-----------------------------------------------------+           |
|                                                                      |
|   CHARACTERISTICS:                                                    |
|   * Very small file size                                             |
|   * Needs synthesizer to play                                        |
|   * Editable at note level                                           |
|   * Used in karaoke, mobile ringtones                                |
|   * Supports 16 channels per device                                  |
|   * 128 standard instruments (GM)                                    |
|                                                                      |
|   MIDI VS AUDIO COMPARISON:                                          |
|   MIDI: +--------------+  +--------------+                           |
|   File  | Instructions|-->| Synthesizer |--> Sound                   |
|         +--------------+  +--------------+                           |
|                                                                      |
|   AUDIO: +--------------+  +--------------+                          |
|   File   | Actual Sound |--> | Speakers    |--> Sound               |
|          | Data         |    |             |                         |
|          +--------------+  +--------------+                          |
|                                                                      |
+----------------------------------------------------------------------+
```

Visual Learning: Flowcharts

Flowchart: Sound Format Selection Guide

```
+-----------------------------------------------------------------+
|             SOUND FORMAT SELECTION GUIDE                         |
+-----------------------------------------------------------------+
+-----------------------------------------------------------------+
| RECOMMENDATION TABLE                                             |
+-----------------------------------------------------------------+
|                                                                   |
| PURPOSE                    RECOMMENDED FORMAT                    |
|------------------------+---------------------------------------|
| Professional Recording  | WAV, FLAC                              |
| Music Listening         | MP3 (192-320 kbps), AAC                |
| Podcasts                | MP3 (128-192 kbps)                     |
| Web Streaming           | MP3 (128 kbps), OGG                    |
| Mobile Devices          | AAC, MP3                               |
| Archiving               | FLAC, ALAC                             |
| Voice Recording         | WAV, MP3 (64 kbps)                     |
| Video Production        | WAV, AIFF                              |
| Ringtone                | MIDI, MP3 (low bitrate)                |
| Karaoke                 | MIDI                                   |
|                                                                   |
+-----------------------------------------------------------------+
```

Flowchart: Audio Editing Workflow

```
+-----------------------------------------------------------------+
|                    AUDIO EDITING WORKFLOW                         |
+-----------------------------------------------------------------+
+------------------+  +------------------+  +------------------+
|    TRIMMING       |  | NOISE REDUCTION |  |  NORMALIZATION    |
|  Remove Silence   |  |   Remove Hiss    |  | Volume Levelling |
|  Cut Sections     |  |   Background     |  | Peak/Average      |
|  Crop Audio       |  |   Filtering      |  | Adjustment        |
+------------------+  +------------------+  +------------------+
+------------------+  +------------------+  +------------------+
|  EQUALIZATION     |  |     EFFECTS      |  |    MULTI-TRACK    |
|  Frequency       |  |     Reverb      |  |     Mixing       |
|  Adjustment      |  |     Echo        |  |    Layering      |
|  Filtering       |  |     Chorus      |  | Synchronization  |
+------------------+  +------------------+  +------------------+
```

Quick Reference Summary

Key Terms

1. **Amplitude** - Volume/loudness of sound
2. **Frequency** - Pitch of sound (Hz)
3. **Sampling Rate** - Samples per second (Hz)
4. **Bit Depth** - Bits per sample (resolution)
5. **Mono** - Single audio channel
6. **Stereo** - Two audio channels
7. **Lossy Compression** - Removes data to reduce size
8. **Lossless Compression** - Maintains quality, reduces size
9. **Psychoacoustics** - Study of sound perception
0. **ADC** - Analog to Digital Conversion

Sound Channels Quick Reference

```
Mono (1.0)      --> Single speaker
Stereo (2.0)    --> L + R speakers
5.1             --> L + R + Center + Surround + Sub
7.1             --> 5.1 + Side speakers
Dolby Atmos     --> Adds overhead channels
```

Sound File Size Formula

File Size = Sample Rate x Bit Depth x Channels x Duration

Questions

Q1: Define sound and list its attributes. Explain each attribute with its measurement unit.

Q2: Differentiate between mono and stereo sound with suitable examples.

Q3: Explain various sound channel configurations (Mono, Stereo, 5.1, 7.1, Dolby Atmos).

Q4: Describe the role and psychological effects of sound in multimedia applications.

Q5: Compare analog and digital sound in detail. Explain the ADC process.

Q6: What is sampling rate and bit depth? Explain their significance with a quality comparison table.

Q7: Explain the WAV file format including its structure and size calculation.

Q8: Describe the MP3 compression process using the psychoacoustic model.

Q9: Compare various sound file formats (WAV, MP3, AAC, FLAC, OGG, WMA, AIFF).

Q10: What is MIDI? Explain how it differs from digital audio formats.

Note: These notes cover Unit II - Sound in Multimedia as per HPU BCA Semester VI curriculum. Review all topics thoroughly for examinations.

— End of Unit II Notes —